



Feature engineering this second channel has proven to be very successful. My precision has roughly increased by 15 points. This has been a good idea altogether. Go through other possible ideas for feature engineering new channels so as to increase the metrics. We will not necessarily implement them all. But I would like you to find research papers, find NASA documents, and other such sources to ideate more feature engineering avenues.

This is a well-researched area. Here is a structured breakdown of every viable channel, ordered by expected impact, with the source rationale for each.

Channels with Strong Research Backing

1. Global Phase-Folded View (Shallue & Vanderburg 2018 — AstroNet)

This is the most replicated channel in the literature. AstroNet's two-input architecture uses a 2001-bin phase-folded global view of the entire orbital period alongside the 201-bin local view (which is essentially what your channel 0 already is). The global view gives the model visibility into the out-of-transit baseline shape, any secondary eclipse at phase 0.5 (which your channel 1 already addresses separately), ellipsoidal variations, and Doppler boosting — all of which are discriminative against eclipsing binaries.^{[1] [2] [3] [4]}

How to build it: Phase-fold the full detrended light curve on the KOI period, bin into 2001 equally spaced phase bins, median-normalise. Resample to 201 points to match your existing shape. This is computationally cheap and only requires period and t_0 from the KOI catalog, which you already have.

Expected impact: High. AstroNet's ablation showed removing the global view degraded accuracy more than removing the local view. The global view is where eclipsing binary signatures live.^[5]

2. Centroid Time Series (Ansdell et al. 2018 — ExoNet)

ExoNet extended AstroNet by adding the **centroid motion time series** as an additional input channel. The centroid records the pixel-level centre of light of the target star across the observation. During a real planetary transit, the centroid does not move — the dip is on-target. During an eclipsing binary contamination or a background eclipsing binary, the centroid shifts toward the contaminating source during the "transit".^{[6] [7] [8] [9]}

ExoNet with centroid channels improved Mean AUC from 94.1% to 95.3% over the base model. The Transformer-based model from 2023 confirmed that centroid information provides the strongest individual contribution to false-positive discrimination.^{[10] [9] [6]}

How to build it: The Kepler pixel-level data products contain the column and row centroid time series (SAP_X, SAP_Y in the .fits files). Extract a 201-cadence window centred on the same t_0 as channel 0. Normalise to zero-mean within the window. This is a single channel carrying 1D centroid displacement. Alternatively encode as two channels (row + column separately), giving you channels 2 and 3.

Expected impact: High, particularly for false-positive suppression — directly improves precision without harming recall.^[11]

3. Odd/Even Transit Depth Difference Channel (NASA DV Pipeline)

This is one of NASA's own official Data Validation tests. The physical logic: for a real planet on a circular orbit, all transits are identical depth. For an eclipsing binary where the secondary eclipse is close in depth to the primary, the period returned by TPS/BLS is half the true binary period. Consequently, odd-numbered and even-numbered "transits" alternate between the primary eclipse (deeper) and the secondary eclipse (shallower).^{[12] [13] [14] [15]}

How to build it: Given a window labeled positive with period P , extract the previous transit at $t_0 - P$ and the next transit at $t_0 + P$. These are the "odd" and "even" neighbours. Compute the depth-normalised difference time series: $(\text{odd_window} - \text{even_window})$ as a 201-cadence channel. For a real planet this should be $\sim$ zero. For an EB it will show a structured residual. For negative windows, fill with zeros.

Expected impact: Medium-high for precision. The test is specifically designed to catch the most common Kepler false positive category.^{[16] [15]}

4. Local Phase-Folded View (Yu et al. 2019, Osborn et al. 2020)

Several papers added a **phase-folded local view** as a separate channel from the raw time-domain window. Phase folding stacks all transits coherently — if the signal is a real periodic transit, the stacked signal improves SNR as $\sqrt{N_{\text{transits}}}$. If it is a single anomalous dip or noise, the folded view shows a much weaker or absent signal. The comparison between the raw single-transit window (channel 0) and the phase-folded stack is discriminative.^[17]

How to build it: Phase-fold all flux data for the star on the KOI period, bin into 201 bins centred on phase 0, median-normalise. This requires the full detrended light curve and the period —

both available in your pipeline. For negative windows from non-planet stars, fold on a random period (same approach as AstroNet's negative training).

Expected impact: Medium. Most useful for short-period planets where many transits are available and the fold is deep. Less useful for long-period planets with few transits — exactly the hard cases your current model struggles with.

5. Weak Secondary / Residual Channel (NASA DV Weak Secondary Test)

This is distinct from your current secondary eclipse channel. NASA's DV pipeline performs the **Weak Secondary Test**: remove the primary transit model, re-apply the whitening filter, and search the residual for any secondary eclipse at any phase. A real planet produces nothing. An EB produces a secondary at phase 0.5 (which your channel 1 handles) but also potentially at non-0.5 phases for eccentric orbits, and the residual depth itself is informative.^[12]

How to build it: After extracting the primary window, fit a simple trapezoidal transit model to the dip, subtract it from the raw window, and pass the residual as a 201-cadence channel. For a real transit, this residual is flat noise. For a systematic or EB, the model fit will be poor and the residual structured. No KOI metadata required beyond what you already have.

Expected impact: Medium. This is an orthogonal signal to the other channels — it catches cases the others miss.

Channels Worth Considering but More Costly to Build

6. Autocorrelation Function (ACF) of the Out-of-Transit Flux

Stellar variability (starspots, pulsations, granulation) creates quasi-periodic patterns in the light curve. The ACF of the out-of-transit flux within the 201-cadence window reveals whether the dip is embedded in a varying star or a quiet star. Transits on highly variable stars are harder to distinguish from variability-induced dips.^[18]

Build cost: Medium. Compute the ACF of the flux excluding the central transit region (e.g., the outer 80 cadences on each side), normalise, and pass as a 201-cadence channel by interpolating the ACF lag axis.

7. Difference Image Centroid Channel (Kepler Pixel-Level)

More precise than the SAP centroid time series. Requires pixel-level Target Pixel Files (TPFs). Compute the difference between in-transit and out-of-transit pixel images — the centroid of the difference image directly localises the source of the brightness change. This is the most powerful single false-positive discriminator in the Kepler DV reports but is computationally expensive.^{[7] [19]}

Build cost: High. Requires TPF files, not just light curves. Significant preprocessing.

Recommended Priority Order

Priority	Channel	Source
1	Global phase-folded view (2001-bin → resample 201)	AstroNet [1] [3]
2	Centroid row/column time series (2 channels)	ExoNet [8] [9] [11]
3	Odd/even transit depth difference	NASA DV [16] [13] [15]
4	Phase-folded local view (stacked SNR)	Yu et al. [17]
5	Residual after transit subtraction	NASA DV [12]
6	Out-of-transit ACF	[18]
7	Difference image centroid	NASA DV [7] [19]

Channels 1 and 2 are the most battle-tested in peer-reviewed literature and directly extend what you have already done. The global view and centroid channels are what separates ExoNet from AstroNet and account for the largest single accuracy jump in the published literature on this exact problem.



1. <https://www.worldscientific.com/doi/full/10.1142/S2251171722500118>
2. <https://arxiv.org/pdf/2105.06292.pdf>
3. <https://academic.oup.com/mnras/article/488/4/5232/5542238>
4. <https://profiles.ucalgary.ca/sites/default/files/2021-09/Exoplanets+Tech+Memo.pdf>
5. <https://arxiv.org/pdf/1810.13434.pdf>
6. <https://academic.oup.com/mnras/article/522/3/3201/7134087>
7. <https://exoplanetarchive.ipac.caltech.edu/docs/KSCI-19093-003.pdf>
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10. <https://arxiv.org/pdf/2304.14283.pdf>
11. https://www.academia.edu/121511875/Scientific_Domain_Knowledge_Improves_Exoplanet_Transit_Classification_with_Deep_Learning
12. https://exoplanetarchive.ipac.caltech.edu/docs/TCERTCompanion_q1_q16.pdf
13. <https://exoplanetarchive.ipac.caltech.edu/docs/DVOnePageSummaryPageCompanion-dr25-V7.html>
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15. <https://ntrs.nasa.gov/api/citations/20180007069/downloads/20180007069.pdf>
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19. https://exoplanetarchive.ipac.caltech.edu/docs/DVSummaryPageCompanion_q1_q16.html

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